Kindworth Dynamic KQ — Technical Report

This technical report presents a comprehensive analysis of the Kindness Quotient (KQ) dynamics for two different organizational contexts: a startup and an enterprise. For each scenario, we define a set of employees and their roles, specify daily Slack/Teams messages, and compute KQ trajectories, alpha matrices, eigenvalues, and key metrics. All numbers are calculated at the start based on the specified messages—no additional assumptions are introduced. A separate image file (`formulas_image.png`) provides the key formulas used in these calculations.

Model and Methodology

We model cultural dynamics as a continuous-time system in which each person's KQ value decays over time and increases when they receive recognition from peers. The core elements are: - **Decay:** Each person has a per-day decay rate λ. A higher λ represents a fast-moving culture (e.g., startups), while a lower λ represents a slower cultural tempo (e.g., enterprises). - **Recognition Impact:** Each Slack/Teams message has a task type with a predefined weight. We scale these weights by a *culture speed multiplier* to reflect how quickly the organization responds to recognition. The sum of these weighted contributions for messages received on a given day forms the daily input *b*. - **Discrete Update:** At the end of each day *d*, the KQ for person *i* is updated according to

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KQ_{i, d+1} = KQ_{i, d} × e^(−λ) + b_{i, d},
```

where *e^(−λ)* captures the natural decay and *b_{i,d}* is the total weighted recognition received by person *i* on day *d*. - **Alpha Matrix:** Across the entire period, we accumulate the weighted contributions from each sender to each recipient into an **alpha matrix** *A*. The largest eigenvalue of *A* gives an indication of how quickly recognition can amplify cultural momentum. A high eigenvalue suggests that recognition has a compounding effect on culture. The formulas underlying this model are provided in `formulas_image.png`.

Recognition Task Weights

We assign weights to recognition task types to quantify their impact on KQ. Higher weights correspond to acts that are more meaningful to the recipient. The table below lists the key task types and their base weights:

Task Type	Weight
Critical Support	5
Bug Prevention	4
Client Impact	6
Thoroughness	4
Process Improvement	3
Performance Fix	4

Task Type	Weight
Crisis Handling	5
Launch Readiness	6
Team Alignment	4
Innovation	10
Mentorship	8
Quick Thanks	3
Data Pipeline Improvement	8
Design Innovation	7
Reliability Enhancement	6
Employee Engagement	5
Customer Renewal	7
Training Session	4
Product Strategy	6
Sales Achievement	7
Budget Optimization	6
Team Collaboration	5

Startup Scenario

Team Overview: A three-person startup consisting of: - Sarah — Senior Product Manager - Alex — Engineering Lead - Maria — QA Lead

The startup operates in a fast-paced environment. We set the decay rate $\lambda = 0.25$/day and apply a culture speed multiplier of 1.2 to all recognition weights. Initial KQ values are 30 for each employee.

Daily Slack/Teams Messages: - **Day 1:** - At 09:00, Sarah recognized Alex for *Critical Support*. - At 14:00, Alex recognized Maria for *Bug Prevention*. - **Day 2:** - At 10:00, Maria recognized Sarah for *Client Impact*. - **Day 3:** - At 09:30, Sarah recognized Maria for *Thoroughness*. - At 13:00, Alex recognized Sarah for *Mentorship*. - **Day 4:** - At 11:00, Alex recognized Sarah for *Process Improvement*. - At 15:00, Maria recognized Alex for *Performance Fix*. - **Day 5:** - At 10:00, Sarah recognized Alex for *Crisis Handling*. - **Day 6:** - At 09:00, Alex recognized Maria for *Launch Readiness*. - At 14:00, Sarah recognized Alex for *Innovation*. - **Day 7:** - At 10:00, Maria recognized Sarah for *Team Alignment*. - At 16:00, Alex recognized Maria for *Quick Thanks*.

Daily KQ and Inputs: | Day | Sarah | Alex | Maria | |—|—|—|—| | 0 | 30.00 | 30.00 | 30.00 | | 1 | 23.36 | 29.36 | 28.16 | | 2 | 25.40 | 22.87 | 21.93 | | 3 | 29.38 | 17.81 | 21.88 | | 4 | 26.48 | 18.67 | 17.04 | | 5 | 20.62 | 20.54 | 13.27 | | 6 | 16.06 | 28.00 | 17.54 | | 7 | 17.31 | 21.80 | 17.26 |

Daily Alpha Inputs (b): | Day | Sarah | Alex | Maria | |—|—|—|—| | 1 | 0.00 | 6.00 | 4.80 | | 2 | 7.20 | 0.00 | 0.00 | | 3 | 9.60 | 0.00 | 4.80 | | 4 | 3.60 | 4.80 | 0.00 | | 5 | 0.00 | 6.00 | 0.00 | | 6 | 0.00 | 12.00 | 7.20 | | 7 | 4.80 | 0.00 | 3.60 |

Alpha Matrix and Eigenvalue: | Recipient Sender | Sarah | Alex | Maria | |—|—|—|—| | Sarah | 0.0 | 13.2 | 12.0 | | Alex | 24.0 | 0.0 | 4.8 | | Maria | 4.8 | 15.6 | 0.0 |

Largest eigenvalue of the alpha matrix: **25.28**

Metrics Summary:

Person | Role | End KQ | Avg KQ | Daily Volatility | Days Under Risk | Time to Risk (days) | Acts Received | Alpha Points | Diversity Index | Unique Peers |

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Sarah	Senior Product Manager	17.31	22.66	4.20	0	0.57	4	25.20	0.18	2
Alex	Engineering Lead	21.80	22.72	5.11	0	1.50	4	28.80	0.18	2
Maria	QA Lead	17.26	19.58	3.53	1	0.56	4	20.40	0.18	2

Enterprise Scenario

Team Overview: Seven employees across multiple departments: - Nina — PMO - Omar — Finance Manager - Liam — Data Engineer - Ava — Design Lead - Ethan — SRE - Priya — HR Manager - Noah — Sales Lead

The enterprise operates with a slower cultural tempo. We set the decay rate $\lambda = 0.15$/day and use a culture speed multiplier of 1.0. Initial KQ values are 30 for each employee.

Daily Slack/Teams Messages: - **Day 1:** - At 09:00, Nina recognized Liam for *Data Pipeline Improvement*. - At 12:00, Liam recognized Ava for *Design Innovation*. - At 15:00, Omar recognized Priya for *Training Session*. - **Day 2:** - At 11:00, Ava recognized Ethan for *Reliability Enhancement*. - At 14:00, Ethan recognized Nina for *Team Collaboration*. - At 16:00, Priya recognized Noah for *Training Session*. - **Day 3:** - At 10:00, Noah recognized Omar for *Budget Optimization*. - At 13:00, Liam recognized Priya for *Training Session*. - At 15:00, Nina recognized Ava for *Design Innovation*. - **Day 4:** - At 09:30, Ethan recognized Liam for *Reliability Enhancement*. - At 13:30, Priya recognized Nina for *Team Collaboration*. - At 16:00, Omar recognized Ethan for *Budget Optimization*. - **Day 5:** - At 10:00, Ava recognized Priya for *Employee Engagement*. - At 14:00, Nina recognized Noah for *Sales Achievement*. - At 16:00, Omar recognized Nina for *Product Strategy*. - **Day 6:** - At 09:00, Priya recognized Ethan for *Training Session*. - At 11:00, Liam recognized Nina for *Mentorship*. - At 15:00, Noah recognized Ava for *Design Innovation*. - **Day 7:** - At 10:00, Ethan recognized Omar for *Budget Optimization*. - At 13:00, Nina recognized Priya for *Mentorship*. - At 16:00, Ava recognized Liam for *Data Pipeline Improvement*. - **Day 8:** - At 09:00, Omar recognized Noah for *Customer Renewal*. - At 12:00, Priya recognized Ava for *Team Collaboration*. - At 15:00, Liam recognized Ethan for *Reliability Enhancement*. - **Day 9:** - At 10:30, Noah recognized Nina for *Product Strategy*. - At 13:30, Ethan recognized Priya for *Employee Engagement*. - At 16:30, Ava recognized Omar for *Budget Optimization*. - **Day**

10: - At 09:30, Nina recognized Ethan for *Reliability Enhancement*. - At 12:30, Priya recognized Liam for *Mentorship*. - At 15:30, Omar recognized Ava for *Design Innovation*.

Daily KQ Values: | Day | Nina | Omar | Liam | Ava | Ethan | Priya | Noah | |—|—|—|—|—|—|—|—| | 0 | 30.00 | 30.00 | 30.00 | 30.00 | 30.00 | 30.00 | 30.00 | | 1 | 25.82 | 25.82 | 33.82 | 32.82 | 25.82 | 29.82 | 25.82 | | 2 | 27.22 | 22.22 | 29.11 | 28.25 | 28.22 | 25.67 | 26.22 | | 3 | 23.43 | 25.13 | 25.06 | 31.31 | 24.29 | 26.09 | 22.57 | | 4 | 25.17 | 21.63 | 27.57 | 26.95 | 26.91 | 22.46 | 19.43 | | 5 | 27.66 | 18.62 | 23.73 | 23.20 | 23.16 | 24.33 | 23.72 | | 6 | 31.81 | 16.02 | 20.42 | 26.97 | 23.93 | 20.94 | 20.42 | | 7 | 27.38 | 19.79 | 25.58 | 23.21 | 20.60 | 26.02 | 17.57 | | 8 | 23.57 | 17.03 | 22.01 | 24.98 | 23.73 | 22.40 | 22.13 | | 9 | 26.28 | 20.66 | 18.95 | 21.50 | 20.43 | 24.28 | 19.04 | | 10 | 22.62 | 17.78 | 24.31 | 25.50 | 23.58 | 20.90 | 16.39 |

Daily Alpha Inputs (b): | Day | Nina | Omar | Liam | Ava | Ethan | Priya | Noah | |—|—|—|—|—|—|—|—| | 1 | 0.00 | 0.00 | 8.00 | 7.00 | 0.00 | 4.00 | 0.00 | | 2 | 5.00 | 0.00 | 0.00 | 0.00 | 6.00 | 0.00 | 4.00 | | 3 | 0.00 | 6.00 | 0.00 | 7.00 | 0.00 | 4.00 | 0.00 | | 4 | 5.00 | 0.00 | 6.00 | 0.00 | 6.00 | 0.00 | 0.00 | | 5 | 6.00 | 0.00 | 0.00 | 0.00 | 0.00 | 5.00 | 7.00 | | 6 | 8.00 | 0.00 | 0.00 | 7.00 | 4.00 | 0.00 | 0.00 | | 7 | 0.00 | 6.00 | 8.00 | 0.00 | 0.00 | 8.00 | 0.00 | | 8 | 0.00 | 0.00 | 0.00 | 5.00 | 6.00 | 0.00 | 7.00 | | 9 | 6.00 | 6.00 | 0.00 | 0.00 | 0.00 | 5.00 | 0.00 | | 10 | 0.00 | 0.00 | 8.00 | 7.00 | 6.00 | 0.00 | 0.00 |

Alpha Matrix and Eigenvalue: | Recipient Sender | Nina | Omar | Liam | Ava | Ethan | Priya | Noah | |—|—|—|—|—|—|—|—| | Nina | 0.0 | 6.0 | 8.0 | 0.0 | 5.0 | 5.0 | 6.0 | | Omar | 0.0 | 0.0 | 0.0 | 6.0 | 6.0 | 0.0 | 6.0 | | Liam | 8.0 | 0.0 | 0.0 | 8.0 | 6.0 | 8.0 | 0.0 | | Ava | 7.0 | 7.0 | 7.0 | 0.0 | 0.0 | 5.0 | 7.0 | | Ethan | 6.0 | 6.0 | 6.0 | 6.0 | 0.0 | 4.0 | 0.0 | | Priya | 8.0 | 4.0 | 4.0 | 5.0 | 5.0 | 0.0 | 0.0 | | Noah | 7.0 | 7.0 | 0.0 | 0.0 | 0.0 | 4.0 | 0.0 |

Largest eigenvalue of the alpha matrix: **26.33**

Metrics Summary:

Person	Role	End KQ	Avg KQ	Daily Volatility	Days Under Risk	Time to Risk (days)	Acts Received	Alpha Points	Diversity Index	Unique Peers
Nina	PMO	22.62	26.10	3.49	0	0.82	5	30.00	0.14	5
Omar	Finance Manager	17.78	20.47	3.25	5	0.00	3	18.00	0.05	3
Liam	Data Engineer	24.31	25.05	4.21	1	1.30	4	30.00	0.14	4
Ava	Design Lead	25.50	26.47	3.79	0	1.62	5	33.00	0.09	5

Person	Role	End KQ	Avg KQ	Daily Volatility	Days Under Risk	Time to Risk (days)	Acts Received	Alpha Points	Diversity Index	Unique Peers
Ethan	SRE	23.58	24.07	3.30	0	1.10	5	28.00	0.14	5
Priya	HR Manager	20.90	24.29	3.19	0	0.29	5	26.00	0.14	5
Noah	Sales Lead	16.39	21.33	3.28	4	0.00	3	18.00	0.14	3

Implementation Details

All calculations were performed in Python using NumPy and pandas. The KQ updates use the specified decay rates and the daily alpha contributions derived from the Slack/Teams logs. The alpha matrix entries were computed by summing all weighted contributions from each sender to each recipient. Eigenvalues were obtained using NumPy's linear algebra package. No additional assumptions were introduced beyond the predefined people, roles, task types, and messages.

Conclusion

The startup scenario shows rapid fluctuations in KQ due to higher decay and a larger culture speed multiplier. Recognition events quickly translate into gains, but KQ also decays swiftly when messages are infrequent. The largest eigenvalue of the startup alpha matrix (≈ 25.28) suggests a high potential for culture amplification: recognition spreads quickly among the small team.

The enterprise scenario exhibits slower decay and a larger network of cross-functional interactions. Even with more messages, the KQ values evolve more gradually, and the larger eigenvalue (≈ 26.33) indicates strong but more distributed cultural momentum. Insights from these simulations can guide leaders on where to reinforce recognition practices, how to tailor decay rates for their context, and how alpha weights influence cultural dynamics.